

Python for Data Science - 2305CS303

Lab - 12

Roll No. : 128

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1. Concatenate DataFrames Row-wise

Create two DataFrames containing student details from two different groups. Use `pd.concat()` with `axis=0` to concatenate the DataFrames row-wise. Display the original DataFrames and the resulting DataFrame.

```
In [1]: import pandas as pd

df1 = pd.DataFrame({
    'RollNo': [101, 102, 103],
    'Name': ['Ravi', 'Amit', 'Neha'],
    'Marks': [85, 78, 90]
})

df2 = pd.DataFrame({
    'RollNo': [104, 105, 106],
    'Name': ['Raj', 'Pooja', 'Karan'],
    'Marks': [75, 88, 82]
})

df3 = pd.concat([df1, df2], axis=0)

print("Group 1:")
print(df1)

print("\nGroup 2:")
print(df2)

print("\nCombined DataFrame:")
print(df3)
```

Group 1:

	RollNo	Name	Marks
0	101	Ravi	85
1	102	Amit	78
2	103	Neha	90

Group 2:

	RollNo	Name	Marks
0	104	Raj	75
1	105	Pooja	88
2	106	Karan	82

Combined DataFrame:

	RollNo	Name	Marks
0	101	Ravi	85
1	102	Amit	78
2	103	Neha	90
0	104	Raj	75
1	105	Pooja	88
2	106	Karan	82

2. Concatenate DataFrames Column-wise

Create two DataFrames containing student personal details and academic details with the same index. Use `pd.concat()` with `axis=1` to combine the DataFrames column-wise. Display the resulting DataFrame.

```
In [4]: df1 = pd.DataFrame({
        'Name': ['Ravi', 'Amit', 'Neha'],
        'Age': [20, 21, 20]
    }, index=[101, 102, 103])

    df2 = pd.DataFrame({
        'Course': ['BCA', 'BCA', 'BCA'],
        'Marks': [85, 78, 90]
    }, index=[101, 102, 103])

    df3 = pd.concat([df1, df2], axis=1)

    print("Combined DataFrame:")
    print(df3)
```

Combined DataFrame:

	Name	Age	Course	Marks
101	Ravi	20	BCA	85
102	Amit	21	BCA	78
103	Neha	20	BCA	90

3. Concatenate DataFrames with Different Columns

Create two DataFrames containing student records with some common and some different columns. Use `pd.concat()` with `axis=0` and observe how pandas handles columns that are not present in both DataFrames.

```
In [6]: df1 = pd.DataFrame({
        'RollNo': [101, 102],
```

```

        'Name': ['Ravi', 'Amit'],
        'Marks': [85, 78]
    })

df2 = pd.DataFrame({
    'RollNo': [103, 104],
    'Name': ['Neha', 'Pooja'],
    'Grade': ['A', 'B']
})

df3 = pd.concat([df1, df2], axis=0)

print(df1)
print(df2)

print("\nCombined DataFrame:")
print(df3)

```

	RollNo	Name	Marks
0	101	Ravi	85
1	102	Amit	78

	RollNo	Name	Grade
0	103	Neha	A
1	104	Pooja	B

Combined DataFrame:

	RollNo	Name	Marks	Grade
0	101	Ravi	85.0	NaN
1	102	Amit	78.0	NaN
0	103	Neha	NaN	A
1	104	Pooja	NaN	B

4. Concatenate DataFrames and Reset the Index

Create two DataFrames representing student records from two classes, each having its own index. Concatenate them row-wise using `pd.concat()` and use `ignore_index=True` to generate a new continuous index.

```

In [7]: df1 = pd.DataFrame({
        'Name': ['Ravi', 'Amit', 'Neha'],
        'Marks': [85, 78, 90]
    }, index=[101, 102, 103])

df2 = pd.DataFrame({
    'Name': ['Raj', 'Pooja', 'Karan'],
    'Marks': [75, 88, 82]
}, index=[201, 202, 203])

df3 = pd.concat([df1, df2], axis=0, ignore_index=True)

print(df1)
print(df2)

```

```
print("\nCombined DataFrame:")
print(df3)
```

	Name	Marks
101	Ravi	85
102	Amit	78
103	Neha	90

	Name	Marks
201	Raj	75
202	Pooja	88
203	Karan	82

Combined DataFrame:

	Name	Marks
0	Ravi	85
1	Amit	78
2	Neha	90
3	Raj	75
4	Pooja	88
5	Karan	82

5. Concatenate Multiple DataFrames

Create three DataFrames containing student records from three semesters. Use

`pd.concat()` to combine all three DataFrames row-wise. Display the final DataFrame and its shape.

```
In [8]: df1 = pd.DataFrame({
        'RollNo': [101, 102],
        'Name': ['Ravi', 'Amit'],
        'Marks': [75, 82]
    })

    df2 = pd.DataFrame({
        'RollNo': [103, 104],
        'Name': ['Neha', 'Pooja'],
        'Marks': [88, 79]
    })

    df3 = pd.DataFrame({
        'RollNo': [105, 106],
        'Name': ['Raj', 'Karan'],
        'Marks': [91, 85]
    })

    final_df = pd.concat([df1, df2, df3], axis=0, ignore_index=True)

    print("Final DataFrame:")
    print(final_df)

    print("\nShape of DataFrame:")
    print(final_df.shape)
```

Final DataFrame:

	RollNo	Name	Marks
0	101	Ravi	75
1	102	Amit	82
2	103	Neha	88
3	104	Pooja	79
4	105	Raj	91
5	106	Karan	85

Shape of DataFrame:

(6, 3)

6. Join Two DataFrames using Index

Create two DataFrames containing student information and marks, with `RollNo` as the index. Use the `join()` method to combine the DataFrames based on their index. Display the resulting DataFrame.

```
In [9]: df1 = pd.DataFrame({
        'Name': ['Ravi', 'Amit', 'Neha'],
        'Course': ['BCA', 'BCA', 'BCA']
    }, index=[101, 102, 103])

    df1.index.name = 'RollNo'

    df2 = pd.DataFrame({
        'Marks': [85, 78, 90],
        'Grade': ['A', 'B', 'A']
    }, index=[101, 102, 103])

    df2.index.name = 'RollNo'
    result = df1.join(df2)

    print(result)
```

	Name	Course	Marks	Grade
RollNo				
101	Ravi	BCA	85	A
102	Amit	BCA	78	B
103	Neha	BCA	90	A

7. Join DataFrames with Different Index Values

Create two DataFrames containing student details and attendance information with partially matching indexes. Demonstrate `join()` using the default left join and then using `how='inner'`. Compare the results.

```
In [10]: df1 = pd.DataFrame({
        'Name': ['Ravi', 'Amit', 'Neha', 'Pooja'],
        'Course': ['BCA', 'BCA', 'BCA', 'BCA']
    }, index=[101, 102, 103, 104])

    df1.index.name = 'RollNo'

    df2 = pd.DataFrame({
        'Attendance': [85, 90, 78]
```

```

}, index=[101, 103, 105])

df2.index.name = 'RollNo'

left_result = df1.join(df2)

inner_result = df1.join(df2, how='inner')

print("Left Join:")
print(left_result)

print("\nInner Join:")
print(inner_result)

```

Left Join:

	Name	Course	Attendance
RollNo			
101	Ravi	BCA	85.0
102	Amit	BCA	NaN
103	Neha	BCA	90.0
104	Pooja	BCA	NaN

Inner Join:

	Name	Course	Attendance
RollNo			
101	Ravi	BCA	85
103	Neha	BCA	90

8. Merge DataFrames using a Common Column

Create two DataFrames containing student details and examination marks. Both DataFrames should contain a common column named `RollNo`. Use `pd.merge()` to combine them based on `RollNo` and display the result.

```

In [12]: df1 = pd.DataFrame({
        'RollNo': [101, 102, 103],
        'Name': ['Ravi', 'Amit', 'Neha'],
        'Course': ['BCA', 'BCA', 'BCA']
    })
    df2 = pd.DataFrame({
        'RollNo': [101, 102, 103],
        'Marks': [85, 78, 90],
        'Grade': ['A', 'B', 'A']
    })

    result = pd.merge(df1, df2, on='RollNo')

    print(result)

```

	RollNo	Name	Course	Marks	Grade
0	101	Ravi	BCA	85	A
1	102	Amit	BCA	78	B
2	103	Neha	BCA	90	A

9. Merge DataFrames using Different Join Types

Create two DataFrames containing student records from two different sources with some common and some non-common `RollNo` values. Use `pd.merge()` to demonstrate

`inner`, `left`, `right`, and `outer` joins. Display and compare all four results.

```
In [13]: df1 = pd.DataFrame({
    'RollNo': [101, 102, 103, 104],
    'Name': ['Ravi', 'Amit', 'Neha', 'Pooja']
})

df2 = pd.DataFrame({
    'RollNo': [103, 104, 105, 106],
    'Marks': [85, 90, 78, 88]
})

inner = pd.merge(df1, df2, on='RollNo', how='inner')

left = pd.merge(df1, df2, on='RollNo', how='left')

right = pd.merge(df1, df2, on='RollNo', how='right')

outer = pd.merge(df1, df2, on='RollNo', how='outer')

print("Inner Join:")
print(inner)

print("\nLeft Join:")
print(left)

print("\nRight Join:")
print(right)

print("\nOuter Join:")
print(outer)
```

Inner Join:

	RollNo	Name	Marks
0	103	Neha	85
1	104	Pooja	90

Left Join:

	RollNo	Name	Marks
0	101	Ravi	NaN
1	102	Amit	NaN
2	103	Neha	85.0
3	104	Pooja	90.0

Right Join:

	RollNo	Name	Marks
0	103	Neha	85
1	104	Pooja	90
2	105	NaN	78
3	106	NaN	88

Outer Join:

	RollNo	Name	Marks
0	101	Ravi	NaN
1	102	Amit	NaN
2	103	Neha	85.0
3	104	Pooja	90.0
4	105	NaN	78.0
5	106	NaN	88.0

10. Combine Student Academic Data using Concat, Join and Merge

Create three DataFrames containing student personal information, semester marks, and attendance information. Use `pd.concat()` to combine two suitable DataFrames, `join()` to combine data using an index, and `pd.merge()` to combine data using a common column. Display the final combined student dataset.

```
In [20]: import pandas as pd

student = pd.DataFrame({
    'RollNo': [101, 102, 103],
    'Name': ['Ravi', 'Amit', 'Neha'],
    'Course': ['BCA', 'BCA', 'BCA']
})

marks = pd.DataFrame({
    'RollNo': [101, 102, 103],
    'Marks': [85, 78, 90],
    'Status': ['Pass', 'Pass', 'Pass']
})

attendance = pd.DataFrame({
    'Attendance': [92, 85, 88],
    'Status': ['Good', 'Good', 'Good']
}, index=[101, 102, 103])

attendance.index.name = 'RollNo'
```

```

newdf1 = pd.concat([student, marks], axis=1)

print("After Concat:")
print(newdf1)

marks1 = marks.set_index('RollNo')

newdf2 = marks1.join( attendance,how="inner", lsuffix="_marks",rsuffix="_attenda

print("\nAfter Join:")
print(newdf2)

newdf3 = pd.merge( student, marks, on='RollNo')

newdf3 = pd.merge( newdf3, attendance.reset_index(), on='RollNo' )

print("\nFinal Combined Student Dataset:")
print(newdf3)

```

After Concat:

	RollNo	Name	Course	RollNo	Marks	Status
0	101	Ravi	BCA	101	85	Pass
1	102	Amit	BCA	102	78	Pass
2	103	Neha	BCA	103	90	Pass

After Join:

	RollNo	Marks	Status_marks	Attendance	Status_attendance
	101	85	Pass	92	Good
	102	78	Pass	85	Good
	103	90	Pass	88	Good

Final Combined Student Dataset:

	RollNo	Name	Course	Marks	Status_x	Attendance	Status_y
0	101	Ravi	BCA	85	Pass	92	Good
1	102	Amit	BCA	78	Pass	85	Good
2	103	Neha	BCA	90	Pass	88	Good

In []: